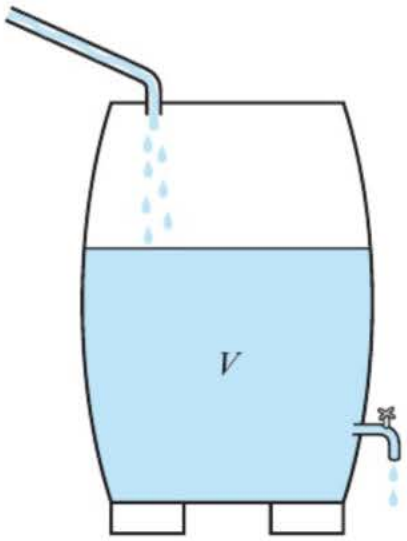
12.11 Modelling with differentiation

You can think of $\frac{dy}{dx}$ as $\frac{\text{small change in } y}{\text{small change in } x}$. It represents the **rate of change** of y with respect to x.

If you replace y and x with variables that represent real-life quantities, you can use the derivative to model lots of real-life situations involving rates of change.



The volume of water in this water butt is constantly changing over time. If V represents the volume of water in the water butt in litres, and t represents the time in seconds, then you could model V as a function of t. If $V = f(t)$ then $\frac{dV}{dt} = f'(t)$ would represent the **rate of change** of volume with respect to time. The units of $\frac{dV}{dt}$ would be litres per second.

Example 21

Given that the volume, V cm^3, of an expanding sphere is related to its radius, r cm, by the formula $V = \frac{4}{3}\pi r^3$, find the rate of change of volume with respect to radius at the instant when the radius is 5 cm.

$V = \frac{4}{3}\pi r^3$

$\frac{dV}{dr} = 4\pi r^2$ — Differentiate V with respect to r. Remember that π is a constant.

When $r = 5$, $\frac{dV}{dr} = 4\pi \times 5^2$ — Substitute $r = 5$.

$= 314$ (3 s.f.)

So the rate of change is 314 cm^3 per cm. — Interpret the answer with units.

Example 22

A large tank in the shape of a cuboid is to be made from 54 m^2 of sheet metal. The tank has a horizontal base and no top. The height of the tank is x metres. Two opposite vertical faces are squares.

a Show that the volume, V m^3, of the tank is given by $V = 18x - \frac{2}{3}x^3$

b Given that x can vary, use differentiation to find the maximum or minimum value of V.

c Justify that the value of V you have found is a maximum.

a Let the length of the tank be y metres.



Total area, $A = 2x^2 + 3xy$

So $54 = 2x^2 + 3xy$

$$y = \frac{54 - 2x^2}{3x}$$

But $V = x^2y$

So $$V = x^2\left(\frac{54 - 2x^2}{3x}\right)$$

$$= \frac{x}{3}(54 - 2x^2)$$

So $$V = 18x - \frac{2}{3}x^3$$

b $$\frac{dV}{dx} = 18 - 2x^2$$

Put $$\frac{dV}{dx} = 0$$

$$0 = 18 - 2x^2$$

So $x^2 = 9$

$x = -3$ or 3

But x is a length so $x = 3$

When $x = 3$, $V = 18 \times 3 - \frac{2}{3} \times 3^3$

$= 54 - 18$

$= 36$

$V = 36$ is a maximum or minimum value of V.

c $$\frac{d^2V}{dx^2} = -4x$$

When $x = 3$, $\frac{d^2V}{dx^2} = -4 \times 3 = -12$

This is negative, so $V = 36$ is the maximum value of V.

Problem-solving

You don't know the length of the tank. Write it as y metres to simplify your working.
You could also draw a sketch to help you find the correct expressions for the surface area and volume of the tank.

Draw a sketch.

Rearrange to find y in terms of x.

Substitute the expression for y into the equation.

Simplify.

Differentiate V with respect to x and put $\frac{dV}{dx} = 0$.

Rearrange to find x.
x is a length so use the positive solution.

Substitute the value of x into the expression for V.

Find the second derivative of V.

$\frac{d^2V}{dx^2} < 0$ so $V = 36$ is a maximum.

Exercise 12K

1 Find $\frac{d\theta}{dt}$ where $\theta = t^2 - 3t$.

2 Find $\frac{dA}{dr}$ where $A = 2\pi r$.

3 Given that $r = \frac{12}{t}$, find the value of $\frac{dr}{dt}$ when $t = 3$.

4 The surface area, $A\,\text{cm}^2$, of an expanding sphere of radius r cm is given by $A = 4\pi r^2$. Find the rate of change of the area with respect to the radius at the instant when the radius is 6 cm.

5 The displacement, s metres, of a car from a fixed point at time t seconds is given by $s = t^2 + 8t$. Find the rate of change of the displacement with respect to time at the instant when $t = 5$.

(P) **6** A rectangular garden is fenced on three sides, and the house forms the fourth side of the rectangle.

a Given that the total length of the fence is 80 m, show that the area, A, of the garden is given by the formula $A = y(80 - 2y)$, where y is the distance from the house to the end of the garden.

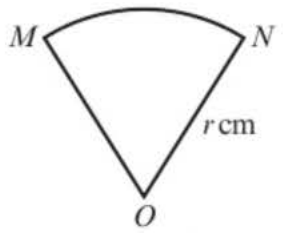
b Given that the area is a maximum for this length of fence, find the dimensions of the enclosed garden, and the area which is enclosed.

(P) **7** A closed cylinder has total surface area equal to 600π.

a Show that the volume, $V\,\text{cm}^3$, of this cylinder is given by the formula $V = 300\pi r - \pi r^3$, where r cm is the radius of the cylinder.

b Find the maximum volume of such a cylinder.

(P) **8** A sector of a circle has area $100\,\text{cm}^2$.

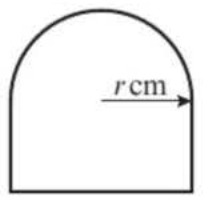


a Show that the perimeter of this sector is given by the formula

$$P = 2r + \frac{200}{r},\ r > \sqrt{\frac{100}{\pi}}$$

b Find the minimum value for the perimeter.

(E/P) **9** A shape consists of a rectangular base with a semicircular top, as shown.



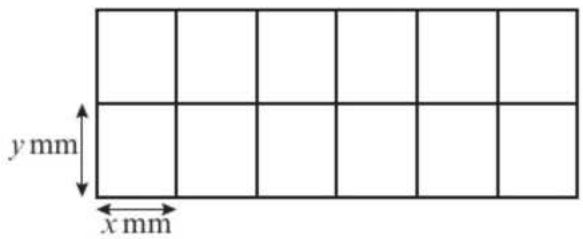
a Given that the perimeter of the shape is 40 cm, show that its area, $A\,\text{cm}^2$, is given by the formula

$$A = 40r - 2r^2 - \frac{\pi r^2}{2}$$

where r cm is the radius of the semicircle. **(2 marks)**

b Hence find the maximum value for the area of the shape. **(4 marks)**

(E/P) **10** The shape shown is a wire frame in the form of a large rectangle split by parallel lengths of wire into 12 smaller equal-sized rectangles.



a Given that the total length of wire used to complete the whole frame is 1512 mm, show that the area of the whole shape, $A\,\text{mm}^2$, is given by the formula

$$A = 1296x - \frac{108x^2}{7}$$

where x mm is the width of one of the smaller rectangles. **(4 marks)**

b Hence find the maximum area which can be enclosed in this way. **(4 marks)**

Mixed exercise 12

E/P **1** Prove, from first principles, that the derivative of $10x^2$ is $20x$. **(4 marks)**

P **2** The point A with coordinates $(1, 4)$ lies on the curve with equation $y = x^3 + 3x$.
The point B also lies on the curve and has x-coordinate $(1 + \delta x)$.

a Show that the gradient of the line segment AB is given by $(\delta x)^2 + 3\delta x + 6$.

b Deduce the gradient of the curve at point A.

3 A curve is given by the equation $y = 3x^2 + 3 + \frac{1}{x^2}$, where $x > 0$. At the points A, B and C on the curve, $x = 1$, 2 and 3 respectively. Find the gradient of the curve at A, B and C.

E **4** Calculate the x-coordinates of the points on the curve with equation $y = 7x^2 - x^3$ at which the gradient is equal to 16. **(4 marks)**

5 Find the x-coordinates of the two points on the curve with equation $y = x^3 - 11x + 1$ where the gradient is 1. Find the corresponding y-coordinates.

E **6** The function f is defined by $f(x) = x + \frac{9}{x}$, $x \in \mathbb{R}$, $x \neq 0$.

a Find $f'(x)$. **(2 marks)**

b Solve $f'(x) = 0$. **(2 marks)**

E **7** Given that

$$y = 3\sqrt{x} - \frac{4}{\sqrt{x}}, \quad x > 0,$$

find $\frac{dy}{dx}$ **(3 marks)**

E/P **8** A curve has equation $y = 12x^{\frac{1}{2}} - x^{\frac{3}{2}}$.

a Show that $\frac{dy}{dx} = \frac{3}{2}x^{-\frac{1}{2}}(4 - x)$. **(2 marks)**

b Find the coordinates of the point on the curve where the gradient is zero. **(2 marks)**

E **9** **a** Expand $(x^{\frac{3}{2}} - 1)(x^{-\frac{1}{2}} + 1)$. **(2 marks)**

b A curve has equation $y = (x^{\frac{3}{2}} - 1)(x^{-\frac{1}{2}} + 1)$, $x > 0$. Find $\frac{dy}{dx}$ **(2 marks)**

c Use your answer to part **b** to calculate the gradient of the curve at the point where $x = 4$. **(1 mark)**

E **10** Differentiate with respect to x:

$$2x^3 + \sqrt{x} + \frac{x^2 + 2x}{x^2}$$ **(3 marks)**

E/P **11** The curve with equation $y = ax^2 + bx + c$ passes through the point $(1, 2)$. The gradient of the curve is zero at the point $(2, 1)$. Find the values of a, b and c. **(5 marks)**